Loan Default Prediction

An interpretable credit-scoring model to approve home equity loans

Capstone Project | Practical Data Science | MIT Professional Education - Applied AI and Data Science Program

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Key takeaways

1 in 5

home equity loans in the data ended in default. Accuracy alone is a misleading yardstick.

62%

default rate when the applicant's debt-to-income ratio is missing, versus 8.6% when it is present. Unverified income is the strongest warning sign.

93%

of defaulters among unseen applicants caught by the proposed Random Forest. 2 of every 3 applicants it flagged really defaulted.

~$9M

indicative annual benefit on 6,000 applications, from roughly $10.8M of credit-error cost down to about $1.9M.

Recommendation: make income and valuation checks mandatory. Deploy the forest at a 0.15 threshold with approve / refer / decline bands, after a shadow run of one to two months and a fairness review.

The problem: approving home equity loans without losing the profit to defaults

- Interest on home loans is a major share of retail bank profit. Defaults consume it.
- Today each decision is manual: slow, inconsistent across underwriters, open to error and bias.
- The consumer credit department wants an empirically derived credit-scoring model built on its own recent loans.
- Constraint: under the Equal Credit Opportunity Act (ECOA) every rejection needs a specific, defensible reason, so the model must be interpretable.

Objective: predict which applicants are likely to default. Tell the bank which attributes to weigh when approving a loan.

The data: Home Equity dataset (HMEQ)

5,960
recent loans

12
attributes known at application

19.9%
defaulted or seriously delinquent

44%
of applications have a blank field

Approach: from raw applications to a decision the bank can defend

1

Explore

Univariate, bivariate and correlation analysis. Default rate by band. Pattern of missing values

2

Prepare

Reserve a stratified 30% test set first. Flag every missing field. Impute median/mode learned on training rows. Cap outliers for the linear model only. One-hot encode

3

Model

Logistic Regression (linear benchmark), Decision Tree, Bagging, Random Forest. Class weights. Cross-validated grid search

4

Measure

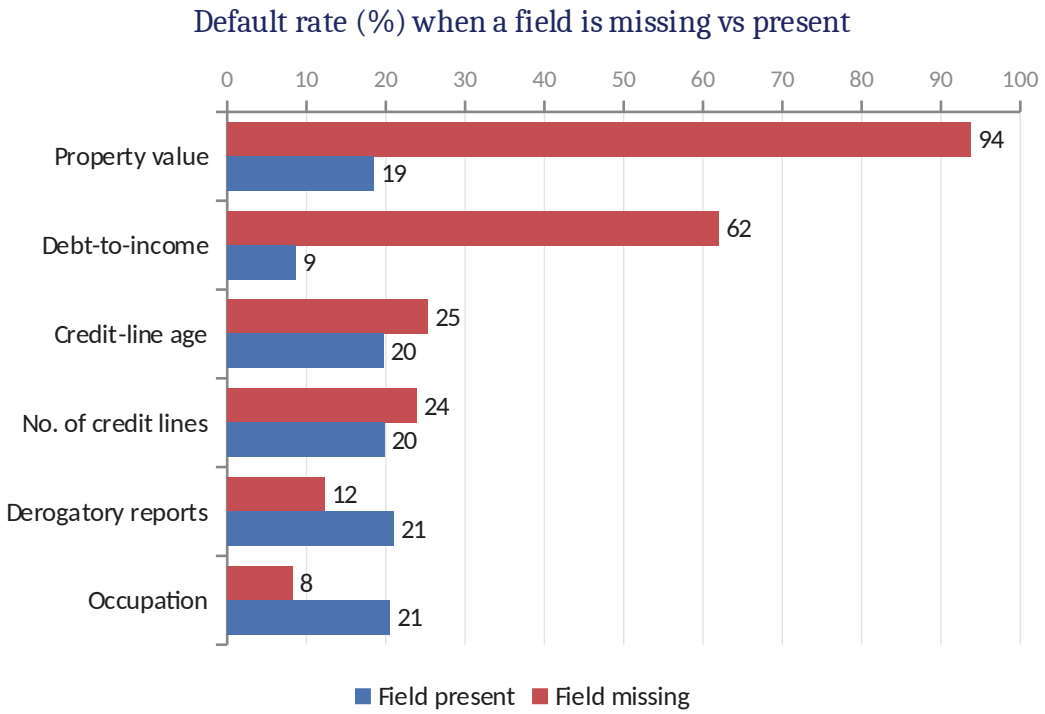
Recall on defaulters as the primary metric. Precision, F1, ROC-AUC as guard-rails. Final model, inputs and threshold chosen on cross-validated expected cost

5

Recommend

Model specification, business rules, implementation plan, benefits, risks

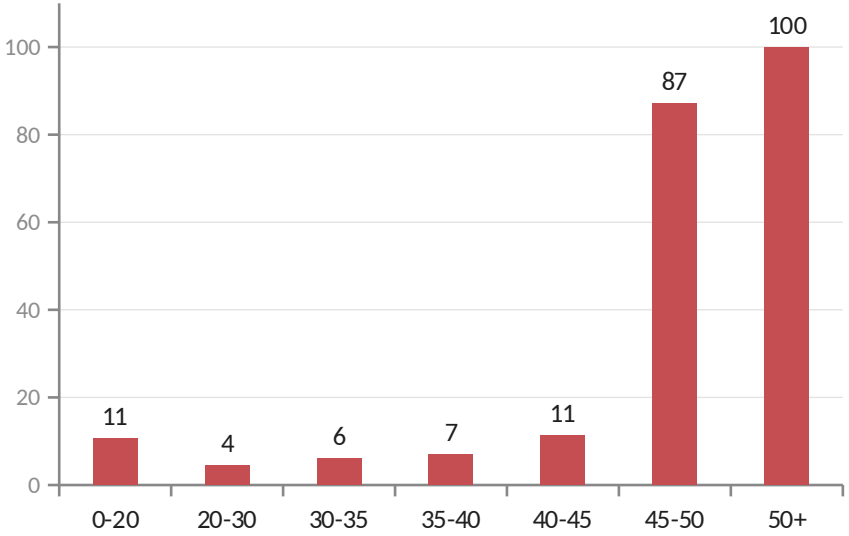
Insight 1: missing information is the strongest warning sign



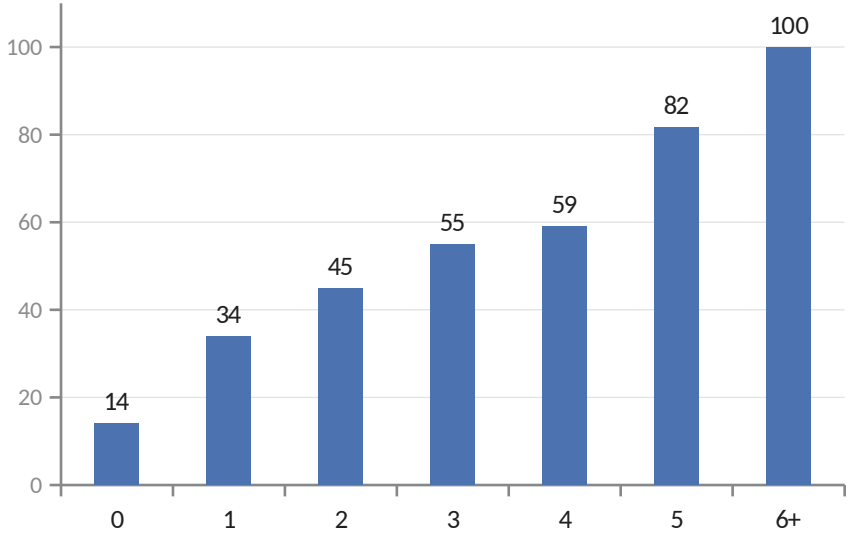
- No recorded debt-to-income ratio: 62% default (1,267 applicants, 21% of the data)
- No property valuation: 94% default (112 applicants)
- Overall default rate 20%: these gaps multiply risk 3 to 5 times, because the bank could not check income or collateral
- Action: a 'was missing' flag for every field with gaps, and mandatory verification in the process

Insight 2: debt burden has a cliff at 45%, and past delinquency repeats

Default rate (%) by debt-to-income ratio band (%)



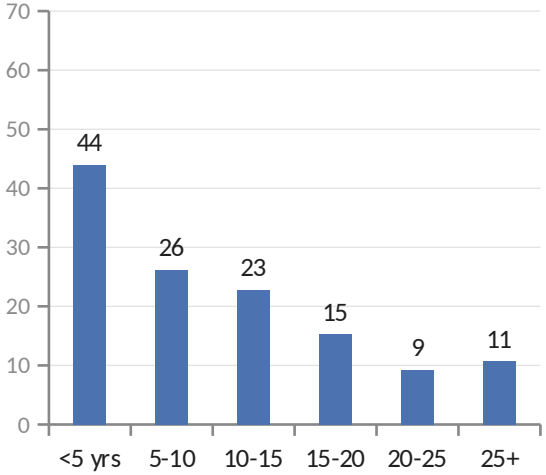
Default rate (%) by delinquent credit lines



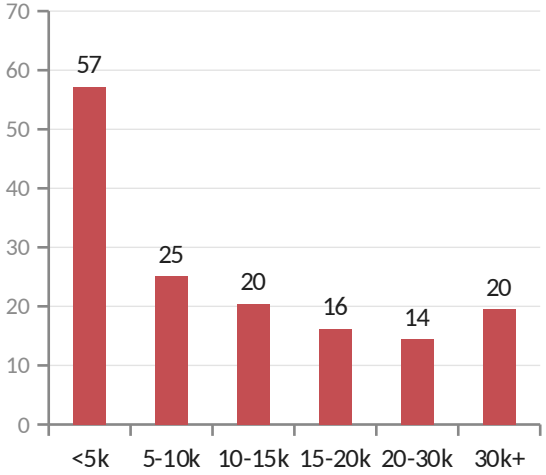
- Debt-to-income ratio 20-40%: only 4-7% default. Above 45%: 87-100%
- Delinquent credit lines: 14% default with none, 34% with one, 100% with six or more

Insight 3: long credit histories protect, small loans do not, occupation matters less

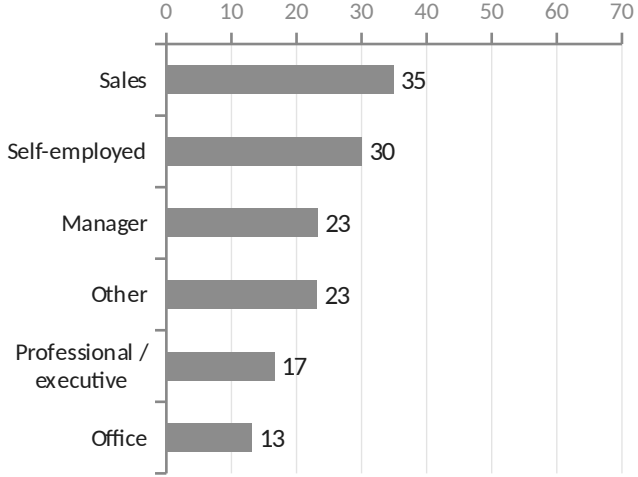
By age of oldest credit line



By loan amount ($)



By occupation



- Credit history under 5 years: 44% default, four times the rate at 20 years or more
- Loans under $5,000: 57% default, the riskiest band
- Occupation: from 13% (office) to 35% (sales), a moderate signal

Measure of success: the two mistakes do not cost the same

Missed defaulter

The model says 'will repay'. The customer defaults.

The bank loses a large part of the principal plus collection cost: about $9,000 on an average $16,900 defaulted loan.

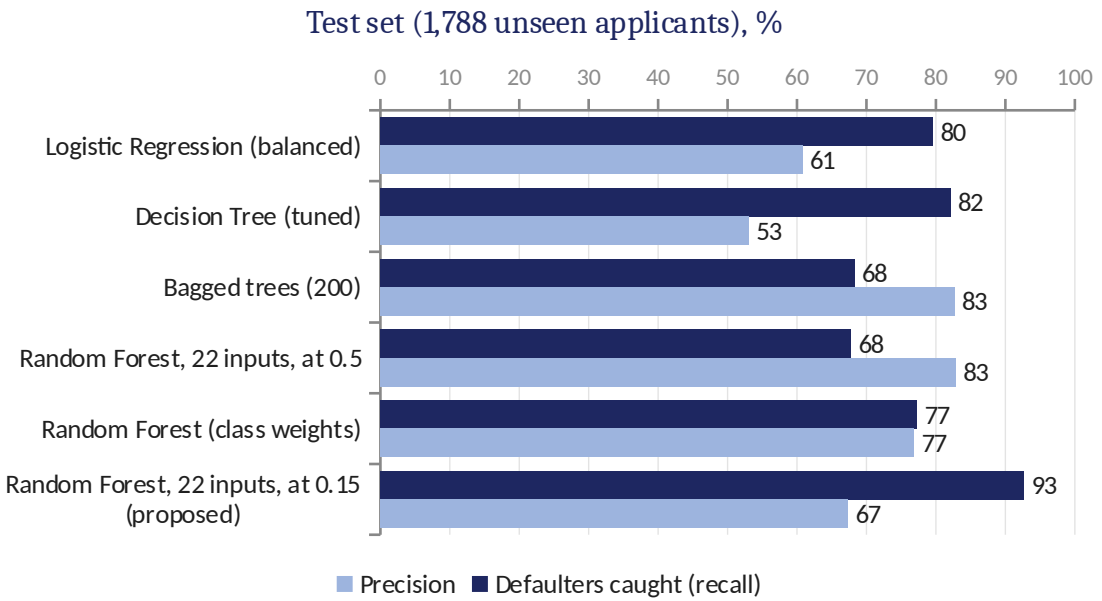
Wrongly declined

The model says 'will default'. The customer is creditworthy.

The bank forgoes the interest margin on one good applicant: about $2,000.

- A missed defaulter costs about 4.5 times a wrongly declined customer
- Primary metric: Recall, the share of future defaulters the model catches
- Guard-rails: Precision (share of flagged applicants who really default), F1 and ROC-AUC (risk-ranking quality)
- Final model and threshold chosen on expected cost, training rows only. Test set scored once

Model comparison: the Random Forest gives the best balance



Expected cost of credit errors per 1,000 applicants, each technique at its own best threshold (out-of-fold, training rows)

Random Forest $324k | Bagged trees $390k | Logistic Regression $550k | Decision Tree $607k

- Logistic regression: 80% caught at 61% precision. A straight line misses the step changes in risk
- Tuned decision tree: 82% caught, but almost one good customer declined per defaulter caught (53% precision)
- Bagged trees and fully grown forests: the best risk ranking (ROC-AUC 0.95 to 0.96)
- Chosen on expected cost, training rows only: plain forest, 22 inputs, threshold 0.15. On test: 93% caught at 67% precision

Proposed solution: a Random Forest scorer explained by a surrogate decision tree

Model specification

- Scoring engine: Random Forest, 200 fully grown trees, no class weights
- Inputs: 22 of 27: 10 numeric attributes, 6 missing-value flags, 6 occupation and loan-purpose indicators
- Threshold: flag above 0.15 (next slide)
- Bands: approve below 0.10 (69% of applicants), refer 0.10-0.25 to an underwriter (7%), decline above 0.25 (24%)
- Explanation: a depth-4 surrogate tree agrees with 93% of the forest's decisions. Its three rules:
  - Debt-to-income ratio missing: flag
  - Ratio above about 44%: flag
  - Otherwise: flag a delinquent credit line with a credit history of about 8 years or less
- Per-applicant risk drivers give each decline its ECOA reason

93%
of defaulters caught

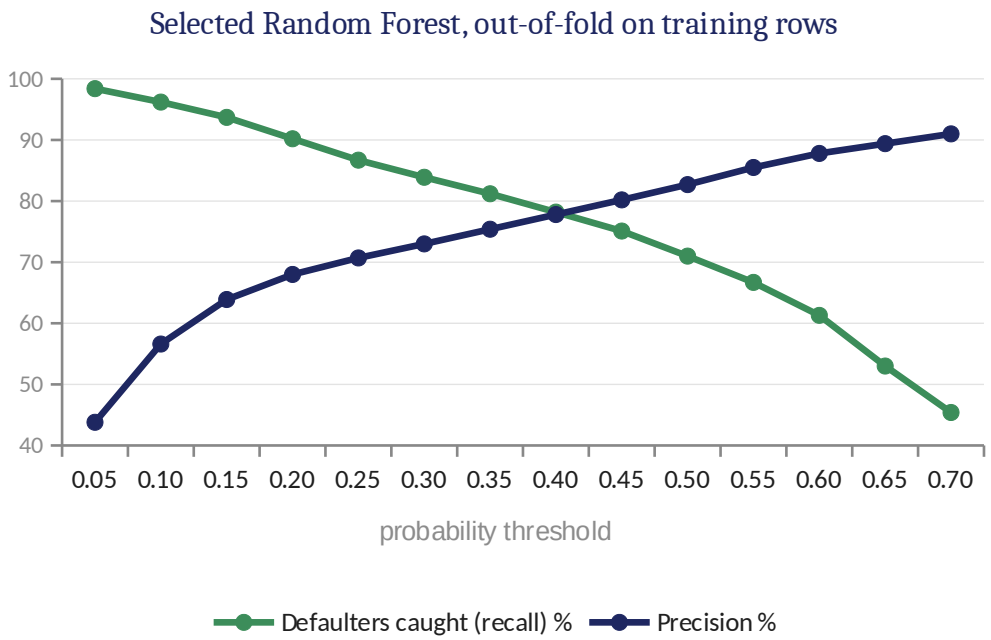
67%
of flagged applicants truly default

0.96
ROC-AUC: risk ranking quality

90%
overall accuracy

At the 0.15 threshold alone, per 1,000 applicants: 185 of 200 defaulters flagged in advance, 15 missed, about 90 good customers flagged.

Why a 0.15 threshold: it is a business dial, not a constant



At 0.50, the usual threshold

- 64 defaulters missed per 1,000
- 28 good customers flagged per 1,000
- $635k indicative cost per 1,000

At 0.15, proposed

- 15 defaulters missed per 1,000
- About 90 good customers flagged per 1,000
- $310k indicative cost per 1,000

Curves: out-of-fold predictions on the training rows. Cards: test set, at $9,000 per missed defaulter and $2,000 per wrongly declined customer. The optimum stays at 0.15 for cost ratios 4.5:1 to 8:1 and on a finer grid.

Business impact: an indicative $9 million a year

Per year, 6,000 applications	Current process	With the proposed model
Loans that default	about 1,200 (20%)	about 90 missed
Default-related losses	about $10.8M	about $0.8M
Good customers flagged at 0.15	none	about 540, margin forgone about $1.1M
Total cost of credit errors	**about $10.8M**	**about $1.9M**

~$9M

indicative gross annual benefit. About $8.8M after running costs, $8.5M in the first year live

- Assumptions: 6,000 applications a year. $9,000 lost per missed defaulter. $2,000 margin forgone per wrongly declined customer. Test performance holds in production
- Measured on the loans the bank approves today, before mandatory verification
- Cost: $300-400k one-off and about $100k a year. Repaid in the first year live even at a quarter of the benefit
- Also: faster, consistent decisions and an auditable reason for every decline

Recommendations and implementation roadmap

1 Months 0-2: prepare

- Mandatory income and valuation verification
- Scoring pipeline and per-applicant risk drivers
- Threshold check with real loss figures. Fairness review

2 Months 2-4: shadow run

- Score every application alongside the manual decision
- Test on past applicants with a recorded ratio. Tune the bands
- Train underwriters on risk drivers and referrals

3 Month 4 onwards: scale

- Automate clear approvals and declines
- Monitor the approval rate monthly, recall and precision as outcomes arrive
- Re-train after 12+ months of verified outcomes, then yearly

Policy rules to adopt now, independent of the model

- Decline or escalate above a 45% debt-to-income ratio (87-100% of these applicants defaulted)
- Refer 2+ delinquent credit lines (45% defaulted at two) or 2+ major derogatory reports (51% at two) to an underwriter
- Treat a missing income or property valuation as high risk, never as neutral
- Where a rule and the score disagree, the stricter outcome applies

Risks, challenges and further analysis

Risks and mitigations

- Selection bias (the data holds approved loans only): shadow run on all applications, then estimate outcomes for declined applicants (reject inference)
- The missing-ratio signal fades once verification is mandatory: re-test on applicants with a recorded ratio before go-live (81% caught at 78% precision)
- Economic drift: monitor monthly, re-train yearly, review the threshold
- Fair lending: 11% of good applicants flagged at 0.15 (3% at 0.5). Disparate-impact test before go-live. Fallback without occupation ready (about 13% more cost)
- Automation bias: per-applicant risk drivers, a referral band, documented overrides

Further analysis

- Reject inference (needs 12+ months of data)
- Engineered features: loan-to-value ratio, debt-to-income bands, interaction terms
- Gradient boosting and probability calibration
- Time-based validation once application dates exist

Thank you

Questions and discussion

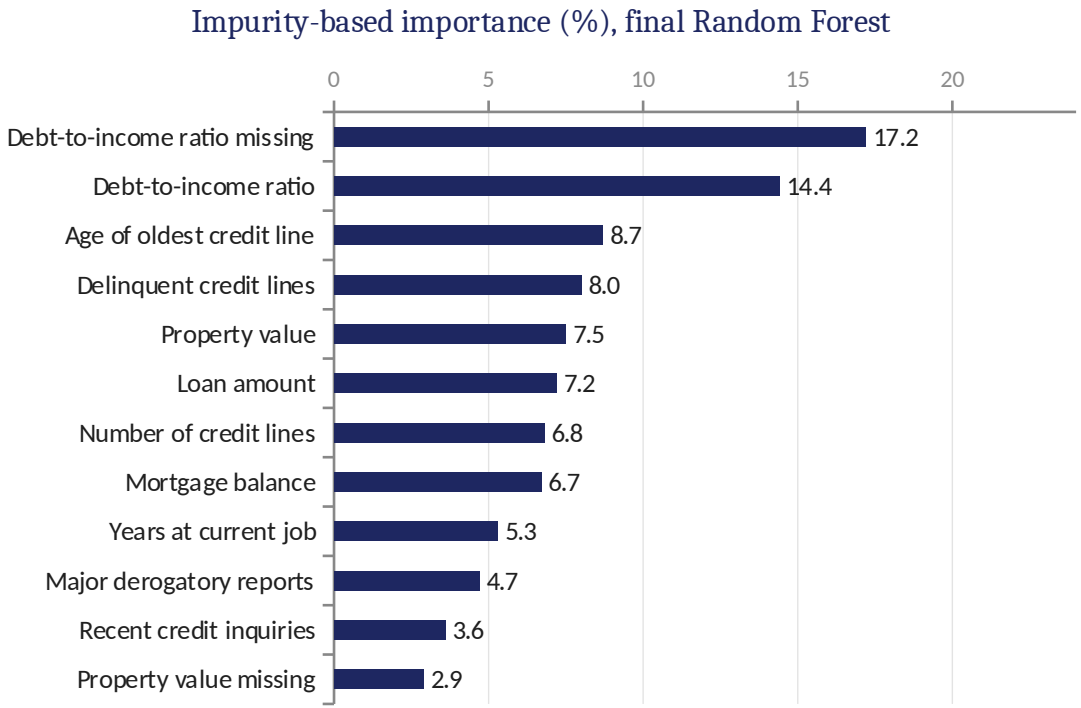
Catch 93% of defaulters. Explain every decision. Save an indicative $9M a year.

Appendix A: all models on the held-out test set

Model	Recall	Precision	F1	Accuracy	ROC-AUC	Train recall
Logistic Regression (plain)	57%	77%	0.65	88%	0.91	63%
Logistic Regression (balanced, regularisation tuned)	80%	61%	0.69	86%	0.91	82%
Decision Tree (standard settings)	69%	71%	0.70	88%	0.81	100%
Decision Tree (class weights 0.2/0.8)	62%	72%	0.66	88%	0.78	100%
Decision Tree (tuned: balanced, at least 20 per leaf)	82%	53%	0.64	82%	0.88	92%
Bagged trees (200 fully grown trees)	68%	83%	0.75	91%	0.95	100%
Bagged trees (class-weighted trees)	60%	81%	0.69	89%	0.95	100%
Random Forest (standard settings, 200 trees)	69%	84%	0.76	91%	0.96	100%
Random Forest (class weights 0.2/0.8)	77%	77%	0.77	91%	0.96	100%
Random Forest (tuned for recall)	79%	66%	0.72	88%	0.93	92%
Random Forest (tuned for recall, 12 inputs)	78%	66%	0.72	88%	0.93	92%
Proposed: Random Forest, 22 inputs, threshold 0.15	**93%**	**67%**	**0.78**	**90%**	**0.96**	-

Test set of 1,788 applicants at the 0.5 threshold except the proposed model. A train recall of 100% is normal for fully grown trees and forests, so only the test columns show how a model generalises. Proposed model: ROC-AUC 0.960 out-of-fold on the training rows, 0.961 on test.

Appendix B: what the Random Forest relies on



- The debt-to-income ratio and its missing flag carry a third of the forest's decision power. With delinquent lines and credit-history length, about half
- Permutation importance ranks the same four at the top, delinquent credit lines first
- Loan, property and mortgage amounts matter in combination, little on their own
- The input-reduction step removed five weak missing flags (about 2% of the decision power) at no cost beyond one standard error
- Logistic-regression coefficients, tree rules and the surrogate tree point to the same drivers

Appendix C: the explanation layer, rules of the surrogate tree that follows the forest

1	**Is the debt-to-income ratio missing?**	Yes: flag, whatever the credit history (62% of these applicants defaulted). Check income before approving.
2	**If recorded, is the ratio above about 44%?**	Yes: flag (87-100% of applicants above 45% defaulted).
3	**Ratio at or below 44%: a delinquent credit line and a credit history of about 8 years or less?**	Yes: flag. Otherwise clear. In this group, applicants with no delinquent line default only about 5% of the time.
4	**How faithful are these rules?**	They reproduce 93% of the forest's decisions on unseen applicants. A rule that never flags anyone would agree 73% of the time. Per-applicant risk drivers give each decline its own reason.
5	**What the rules do not use**	Loan size, derogatory reports, occupation and loan purpose do not change the surrogate's outcome. They matter only in combination inside the forest.

Surrogate: depth-4 tree, at least 20 applicants per leaf, fitted to the forest's out-of-fold flag/clear decisions at threshold 0.15 on the training rows. The stand-alone tuned tree (test recall 82%, precision 53%) found the same drivers.

Appendix D: data treatment and pre-processing

Step	What we did	Why
Duplicates	None found. No rows removed	All 5,960 loans kept
Missing values	A 0/1 'was missing' flag per field, then median (numeric) or mode (categorical) imputation learned on the training rows	A missing debt-to-income ratio or property value is the strongest default signal. Removing rows would lose 44% of the data
Outliers	Inter-quartile capping for the linear model only. Derogatory reports and delinquent lines capped at the 99th percentile. Tree models use raw values	Extreme values distort a linear model (debt-to-income 203%, credit history 97 years). Trees split on rank order
Encoding and scaling	One-hot encoding of loan purpose and occupation (first level dropped). Standard scaling for the linear model only. Flags stay 0/1	Comparable coefficients for the linear model
Split	70% train / 30% test, stratified on the target, fixed seed, reserved before we fitted any statistic	No leakage from test rows into medians, caps or scaling
Imbalance	Class weights inside the individual models, then the decision threshold for the final model. No resampling	Keeps every real applicant. Class weights added nothing once we chose the threshold on cost
Model inputs	22 of 27 columns: the 10 numeric attributes, 6 missing flags (property value, loan purpose, occupation, years at job, derogatory reports, debt-to-income) and 6 loan-purpose and occupation indicators	The input-reduction step removed five flags (mortgage balance, delinquent lines, credit-line age, inquiries, number of credit lines). A final check compared 22 inputs against all 27